



**INTERNATIONAL CAMPUS
OF SCIENCE AND TECHNOLOGY**

International Campus of Science and Technology

Faculty of Computing, Department of IT
HDIT Batch 01

HDIT - DBMS

Project Name: Restaurant Management System - Cafulon

Submitted by:

1. NM. Roshaan Akther
2. A. Washid
3. MAM. Masheef
4. MIF. Fahmidha
5. MNF. Nuha
6. WMCS. Weerakkodi

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USER MANUAL

Cafulon Restaurant Management System

1. Prerequisites

Software	Minimum Version	Download
Python	3.10+	python.org
MySQL	8.0+	Included with XAMPP
XAMPP	8.x	apachefriends.org
Web Browser	Chrome/Firefox/Edge (latest)	Search Web

2. Installation Guide

Step 1: Install XAMPP (MySQL)

1. Download XAMPP from apachefriends.org
2. Run the installer and select MySQL and phpMyAdmin components
3. Launch XAMPP Control Panel and click Start on the MySQL module
4. Verify MySQL is running on port 3306

Step 2: Install Python Dependencies

Open a terminal in the project root directory:

```
pip install -r webapp/requirements.txt
```

This installs Flask 3+ and PyMySQL 1.0+.

Step 3: Create the Database

Option A — Using MySQL Command Line:

```
mysql -u root -p < restaurant_schema.sql
```

Option B — Using phpMyAdmin:

1. Open browser and go to <http://localhost/phpmyadmin>
2. Click Import tab
3. Click Choose File and select `restaurant_schema.sql`
4. Click Go

The script will:

- Create the RestaurantDB database
- Create all 9 tables with constraints
- Insert 24+ sample records per main table
- Create 4 views and 1 stored procedure
- Create 5 indexes

Step 4: Configure Database Credentials

Edit `webapp/app.py` if your MySQL credentials differ:

```
DB = {  
    'host': 'localhost',  
    'user': 'root',  
    'password': 'R00t@12345', # Change to your MySQL password  
    'database': 'RestaurantDB',  
    'cursorclass': pymysql.cursors.DictCursor  
}
```

Step 5: Start the Web Application

```
cd webapp
python app.py
```

The application will start at `http://0.0.0.0:5000`. Open your browser and navigate to `http://localhost:5000`.

3. Application Walkthrough

3.1 Dashboard

The dashboard displays:

- 8 Stat Cards: Total customers, employees, menu items, orders, reservations, revenue, pending orders, today's reservations
- Recent Orders Table: Latest 5 orders with status, amount, and date

3.2 Managing Customers

View Customers:

- Click Customers in the sidebar
- Table shows ID, Name, Phone, Email, City

Add Customer:

- Click the + button in the top bar
- Fill in: First Name, Last Name, Phone, Email, Address, City
- Click Add

Edit Customer:

- Click the pencil icon next to a customer
- Modify fields and click Update

Delete Customer:

- Click the trash icon next to a customer
- Confirm deletion

3.3 Managing Employees

View Employees:

- Click Employees in the sidebar
- Table shows ID, Name, Position, Phone, Salary, Shift

Add Employee:

- Click + in the top bar
- Fill in all required fields and select Position/Shift from dropdowns
- Click Add

3.4 Managing Menu Items

View Menu:

- Click Menu in the sidebar
- Table shows items grouped by category with price and availability

Add Menu Item:

- Click + in the top bar
- Select Category, enter Name, Description, Price, and Preparation Time
- Click Add

3.5 Managing Reservations

View Reservations:

- Click Reservations in the sidebar
- Table shows customer, date, time, party size, table, and status

Add Reservation:

- Click + in the top bar
- Select Customer, Table, Employee (optional)
- Enter Date, Time, Party Size, Status, and Special Requests

- Click Add

3.6 Processing Orders

View Orders:

- Click Orders in the sidebar
- Table shows order ID, customer, employee, type, amount, and status

Create New Order:

1. Click + in the top bar
2. Select Customer (optional — walk-in if empty)
3. Select Employee (required)
4. Select Table (optional)
5. Choose Order Type (Dine-In / Takeaway / Delivery)
6. Add items by selecting from dropdown and entering quantity
7. Click + to add more rows
8. Subtotal and grand total update automatically
9. Click Create

Update Order Status:

1. Click View on an order
2. Use the status dropdown in the right panel
3. Click the refresh button to update

3.7 Recording Payments

View Payments:

- Click Payments in the sidebar

Record Payment:

1. Click + in the top bar
 2. Select an unpaid completed order
 3. Enter Amount and Payment Method
 4. Enter Transaction Reference (optional)
 5. Click Record
-

4. Sample Queries

Basic Queries

```
-- View all available menu items
SELECT ItemName, Price FROM MenuItems WHERE IsAvailable = TRUE;

-- Find a customer by phone number
SELECT * FROM Customers WHERE Phone = '071-2345678';

-- Count total orders by type
SELECT OrderType, COUNT(*) FROM Orders GROUP BY OrderType;
```

Reports

```
-- Daily sales summary
SELECT DATE(OrderDate) AS Day,
       COUNT(*) AS Orders,
       SUM(TotalAmount) AS Revenue
FROM Orders
WHERE Status = 'Completed'
GROUP BY DATE(OrderDate);

-- Top 5 best-selling items
SELECT mi.ItemName, SUM(od.Quantity) AS TotalSold
FROM OrderDetails od
JOIN MenuItems mi ON od.ItemID = mi.ItemID
GROUP BY mi.ItemID
ORDER BY TotalSold DESC
LIMIT 5;

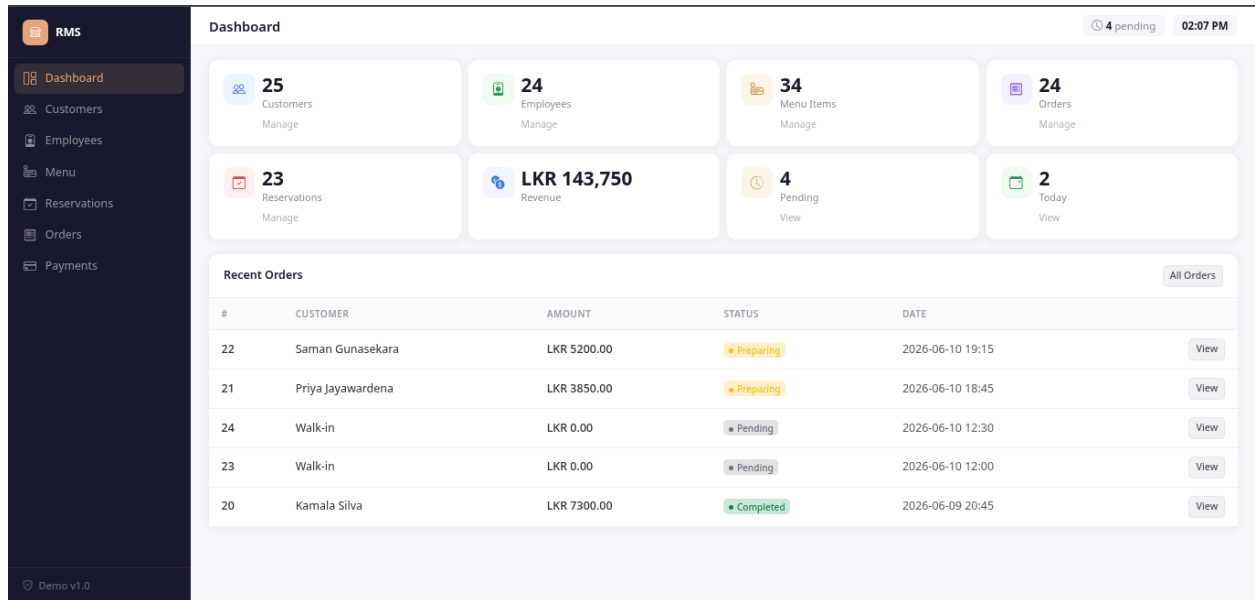
-- Employee performance
SELECT CONCAT(FirstName, ' ', LastName) AS Employee,
       Position, COUNT(o.OrderID) AS OrdersServed
FROM Employees e
LEFT JOIN Orders o ON e.EmployeeID = o.EmployeeID
GROUP BY e.EmployeeID
ORDER BY OrdersServed DESC;
```

5. Troubleshooting

Problem	Solution
Can't connect to MySQL server	Ensure MySQL is running in XAMPP Control Panel
Access denied for user 'root'	Check password in app.py DB config
Table doesn't exist	Run restaurant_schema.sql to create all tables
Port 5000 already in use	Change port in app.py: app.run(port=5001)
ModuleNotFoundError	Run pip install -r requirements.txt

6. Screenshots

Dashboard



Main dashboard showing key metrics and recent orders

Customers List

The Customers List page displays a table of customer information. The table has columns for ID, Name, Phone, Email, and City. Each row includes edit and delete icons. The sidebar on the left shows the navigation menu, and the top right corner displays the time as 02:08 PM.

ID	NAME	PHONE	EMAIL	CITY	
25	Supun Weerasinghe	071-8889990	supun.w@email.com	Colombo	
24	Hasan Rafeek	070-5678900	hasan.r@email.com	Kandy	
23	Sithy Zareena	078-4567899	sithy.z@email.com	Colombo	
22	Ahamed Rizwan	072-3456788	ahamed.r@email.com	Colombo	
21	Fathima Nusrath	071-2345677	fathima.n@email.com	Colombo	
20	Mohamed Ismail	076-1234566	mohamed.i@email.com	Kandy	
19	Deepa Rajendran	077-0123455	deepa.r@email.com	Colombo	
18	Yogeswaran Mahendran	070-9012344	yogeswaran.m@email.com	Batticaloa	
17	Kogulakrishnan Tharmalingam	078-8901233	kogula.t@email.com	Galle	
16	Tharmini Sivapragasam	075-7890122	tharmini.s@email.com	Kandy	
15	Sivakumar Kandasamy	072-6789011	sivakumark@email.com	Jaffna	
14	Wasantha Kumari	070-5678901	wasantha.k@email.com	Badulla	

Customer management screen with edit/delete actions

Menu Items

ID	NAME	CATEGORY	PRICE	PREP	AVAIL		
						EDIT	DELETE
3	Chicken Curry Puffs	Appetizers	LKR 900.00	15m	Yes		
2	Fish Cutlets	Appetizers	LKR 1200.00	12m	Yes		
5	Garlic Prawns	Appetizers	LKR 1800.00	15m	Yes		
1	Pol Sambol with Hoppers	Appetizers	LKR 850.00	10m	Yes		
4	Vegetable Spring Rolls	Appetizers	LKR 750.00	10m	Yes		
33	Ceylon Tea	Beverages	LKR 350.00	5m	Yes		
32	Fresh Mango Juice	Beverages	LKR 550.00	5m	Yes		
31	King Coconut Water	Beverages	LKR 450.00	3m	Yes		
34	Mineral Water	Beverages	LKR 250.00	2m	Yes		
22	Chicken Cheese Burger	Burgers & Sandwiches	LKR 1800.00	15m	Yes		
23	Fish Burger	Burgers & Sandwiches	LKR 1600.00	15m	Yes		
30	Fresh Fruit Platter	Desserts	LKR 900.00	5m	Yes		

Menu items with category, price, and availability

Reservations

RMS

Dashboard

Customers

Employees

Menu

Reservations

Orders

Payments

Demo v1.0

Reservations

02:08 PM

#	CUSTOMER	DATE	TIME	PARTY	TABLE	STATUS	
23	Thusitha Fernando	2026-06-12	19:30:00	4	#15	Confirmed	
22	Shanthi Wickramasinghe	2026-06-11	20:00:00	6	#6	Confirmed	
21	Ruwan Dissanayake	2026-06-11	19:00:00	8	#13	Confirmed	
20	Saman Gunasekara	2026-06-10	19:00:00	2	#2	Confirmed	
19	Priya Jayawardena	2026-06-10	18:30:00	3	#9	Confirmed	
18	Kamala Silva	2026-06-09	20:30:00	6	#7	Seated	
17	Nimal Perera	2026-06-09	19:00:00	5	#5	Seated	
16	Tharmini Sivapragasam	2026-06-08	20:00:00	4	#4	Completed	
15	Sivakumar Kandasamy	2026-06-08	19:30:00	4	#14	Completed	
14	Wasantha Kumari	2026-06-07	19:00:00	2	#1	Completed	
13	Palitha Senanayake	2026-06-07	18:00:00	3	#10	Completed	
12	Sriyani Herath	2026-06-06	20:30:00	8	#8	Completed	

Reservation list with status badges

New Order Form

Orders						02:09 PM	
#	CUSTOMER	EMPLOYEE	TYPE	AMOUNT	STATUS		
22	Saman Gunasekara	Niroshan Fernando	Dine-In	LKR 5200.00	Preparing		
21	Priya Jayawardena	Sampath Jayasinghe	Dine-In	LKR 3850.00	Preparing		
24	Walk-in	Thusitha Kumara	Delivery	LKR 0.00	Pending		
23	Walk-in	Saman Weerawansa	Takeaway	LKR 0.00	Pending		
20	Kamala Silva	Thusitha Kumara	Dine-In	LKR 7300.00	Completed		
19	Nimal Perera	Saman Weerawansa	Dine-In	LKR 6300.00	Completed		
18	Walk-in	Niroshan Fernando	Delivery	LKR 2550.00	Completed		
17	Walk-in	Sampath Jayasinghe	Takeaway	LKR 4050.00	Completed		
16	Tharmini Sivapragasam	Thusitha Kumara	Dine-In	LKR 4550.00	Completed		
15	Sivakumar Kandasamy	Saman Weerawansa	Dine-In	LKR 5800.00	Completed		
14	Wasantha Kumari	Sampath Jayasinghe	Dine-In	LKR 4750.00	Completed		
13	Palitha Senanayake	Saman Weerawansa	Dine-In	LKR 8950.00	Completed		

Order creation with dynamic item rows and auto-calculated totals

Order Detail

Order #22					02:09 PM		
Customer Saman Gunasekara		Employee Niroshan Fernando		Table 2			
Type Dine-In		Date 2026-06-10 19:15		Status Preparing			
Change Status					Preparing		
Items							
ITEM	CATEGORY	PRICE	QTY	SUBTOTAL			
Total				LKR 5200.00			
Payments							
ID	DATE	METHOD	REF	AMOUNT			
No payments							
← Back							

Order view with items, payments, and status changer

OTHER SCREENSHOTS ARE UPLOADED TO THE GITHUB